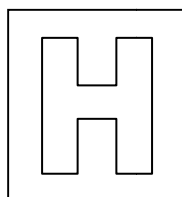


Candidate Name: _____

Class Adm No

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2015 Preliminary Examination 2

Pre-University 3

H2 Biology

9648/03

Applications Paper and Planning Question

21 September 2015

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.
You will lose marks if you do not show your working or if you do not use appropriate units.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This question paper consists of 20 printed pages including 3 blank pages

[Turn over

Answer **all** questions.

1. Spiders cannot be farmed, in contrast to silkworms, due to their aggressive behaviour and territorial nature. Collecting silk from webs is a time-consuming task. It took eight years to make a golden spider silk cape from 1.2 million golden orb webs. Therefore, biotechnological production of recombinant spider silks is the only practicable solution to harvest silks on a larger scale and to meet growing needs of medicine.

A group of scientists tried to clone the *MaSp1*, which codes for the main protein (Spidroin) in the spider web silk, into *Escherichia coli*. Several restriction enzymes were considered for this experiment. The target sites for the restriction enzyme HaeIII, BamHI, HindIII and EcoRI are shown in Fig. 1.1. The arrows drawn in each sequence show where the enzyme cuts the DNA molecule.

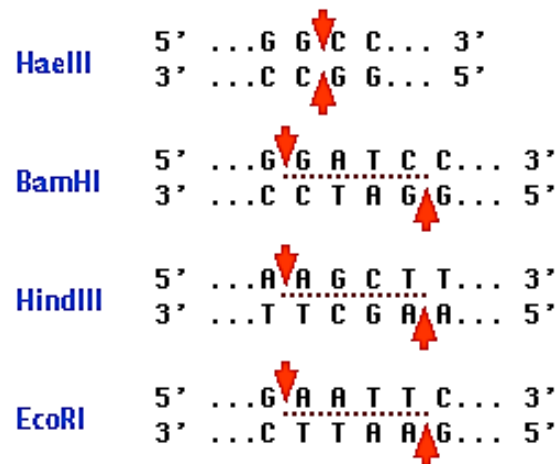


Fig. 1.1

- (a) Explain why it is more suitable to use EcoRI than HaeIII for cloning.

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.....[2]

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Their choice of plasmid is pGEM-7zf(+) vector which is commercially available. The multiple cloning sites (MCS) contain a number of restriction enzyme sites that can be utilised for cloning. Fig. 1.2 shows the restriction map of pGEM-7zf(+) plasmid.

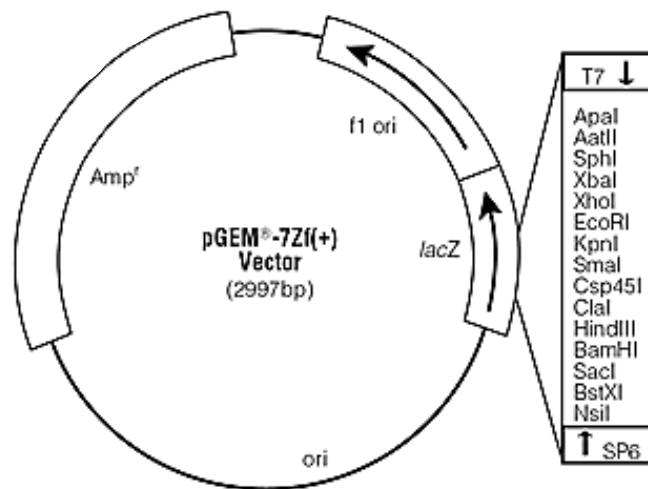


Fig. 1.2

Fig. 1.3 shows the base sequence in the MCS of the plasmid. Apart from the DNA sequence shown, there are no target sequences for BamHI, HindIII and EcoRI in the rest of the plasmid.

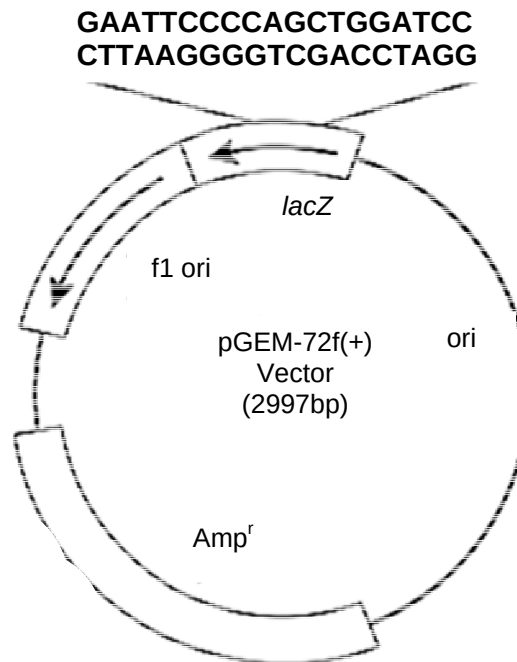


Fig. 1.3

- (b) With reference to Fig. 1.1 and Fig. 1.3, state the number of DNA fragments present after the plasmid has been cut with both BamHI and EcoRI together.

.....[1]

To form recombinant DNA molecule, pGEM-7zf(+) and *MaSp1* were cleaved with both BamHI and EcoRI.

(c)

- (i) Explain what is meant by recombinant DNA molecule.

.....

.....[1]

- (ii) Suggest one reason why different restriction enzymes were used to create different sticky ends on each end of the *MaSp1* gene.

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.....[1]

- (iii) Describe the subsequent steps involved in the formation of recombinant plasmid.

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.....[2]

- (d) Explain how the scientists can use *lacZ* gene to select for bacteria containing the recombinant plasmid.

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.....[3]

The scientists also explored the use of Polymerase Chain Reaction (PCR) to amplify the *MaSpl* gene. A pair forward and reverse primer was designed to amplify the *MaSpl* gene. Excess primers were added and the amplification of the target gene took place in vitro using a thermal cycler.

(e)

(i) Describe how PCR amplifies the *MaSpl* gene.

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.....[3]

(ii) Suggest why excess primers are added to the PCR mixture.

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.....[1]

(iii) Explain why a pair of primers is needed for PCR.

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.....[2]

[Total: 16]

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2. Cystic fibrosis (CF) used to be considered a fatal disease of childhood. With improved diagnosis and treatments to manage the disease, many people with CF now live well into adulthood.

CF is caused by mutation in the *CFTR* gene. The gene of interest can be found on the long arm of chromosome 7. Several restriction fragment length polymorphism (RFLP) markers located on the long arm of chromosome 7 have been found to be closely linked to CF.

One particular RFLP marker is shown in Fig. 2.1. The target sites of *EcoRI* are marked by arrows and the length of DNA between the targets sites are given in kilobases (kb). Most CF patients have a point mutation, which substitutes one of the bases at the restriction site marked with an asterisk (*).

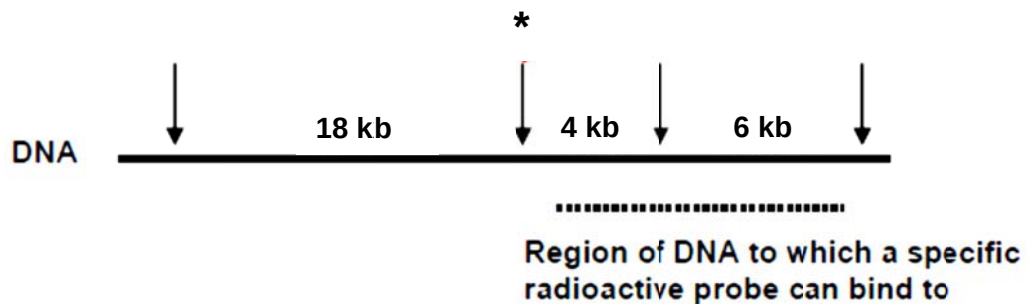


Fig. 2.1

CF can be diagnosed by cutting the DNA from patients with *EcoRI* and separate the DNA fragments by gel electrophoresis. The DNA fragments can then be transferred to nitrocellulose membrane and stained by a radioactive probe.

Fig. 2.2 is an autoradiograph which shows how CF can be detected by the altered size of the DNA fragments produced by cleavage with *EcoRI*. The banding patterns produced by DNA from three individuals, A to C are shown.

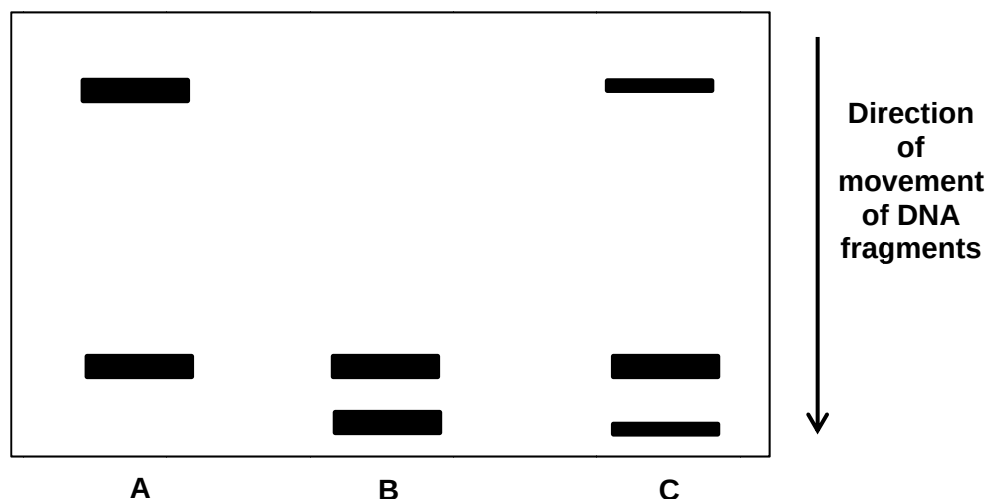


Fig. 2.2

(a)

(i) Identify the DNA fragment of the mutant allele by circling the appropriate band(s) in Fig. 2.2. [1]

(ii) State the size of the DNA fragment identified in **(a)(i)**.

.....[1]

(iii) With reference to Fig. 2.1 and 2.2, explain the banding pattern of individual **C**.

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.....[3]

(b) Suggest why a person without CF can also have the diseased RFLP marker.

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.....[1]

CF is an ideal candidate for gene therapy. Experimental trials have revealed that Lentivirus vectors of the retrovirus family show interesting properties for gene therapy. Lentivirus has a single stranded RNA genome with a reverse transcriptase enzyme.

(c) Describe two advantages of using Lentivirus vector in gene therapy.

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.....[2]

- (d)** Describe the problems that may arise when a viral vector is used to introduce a functional allele into the target cell.

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.....[2]

Severe combined immunodeficiency (SCID) is another ideal candidate for gene therapy. SCID is a potentially fatal primary immunodeficiency. There are at least 13 different genetic defects that can cause SCID. The most common form of SCID, affecting nearly 45% of all cases, is due to a mutation in a gene on the X chromosome. This form of SCID is also known as X-linked SCID.

- (e)** Explain why X-linked SCID is a suitable choice for gene therapy.

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.....[2]

- (f)** Describe the genetic basis of X-linked SCID.

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.....[2]

A group of X-linked SCID patients underwent ex vivo retrovirus-mediated transfer of functional gene into CD34⁺ bone marrow cells. Different dosages of viral vectors were administered for different patients. The bone marrow cells were then transferred back into the patients.

Efficacy of the gene transfer was determined by determining the presence of mRNA of the functional gene one week after the treatment. The results are shown in Table 2.1.

Table 2.1

Patient	Gender	Age / years	Dose / pfu	Presence of mRNA of the functional gene	
				Treated with empty viral vector	Treated with modified viral vector
1	F	25	2×10^7	No	No
2	M	44	2×10^7	No	No
3	M	23	2×10^8	No	No
4	M	21	2×10^8	No	No
5	F	34	2×10^9	No	Yes
6	M	40	2×10^9	No	No
7	F	19	2×10^9	No	Yes
8	F	33	2×10^{10}	No	Yes
9	M	22	2×10^{10}	No	No

(g) Describe the relationship between the vector dosage and treatment efficacy.

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.....[2]

[Total: 16]

3. Iron deficiency is the most common nutritional deficiency in the world. Use of genetic engineering to increase the iron content of the crops can be achieved by the introduction of genes coding for iron-binding proteins and iron-storage proteins.

FvC5sdp is an iron-binding protein expressed in fungus. Five transgenic plants (D-3, D-7, D-2, D-11 and D-12) expressing the gene coding for FvC5sdp were created. The total iron content of the fruits from wild type control and the transgenic lines were measured by atomic absorption spectroscopy. The results are shown in Fig. 3.1.

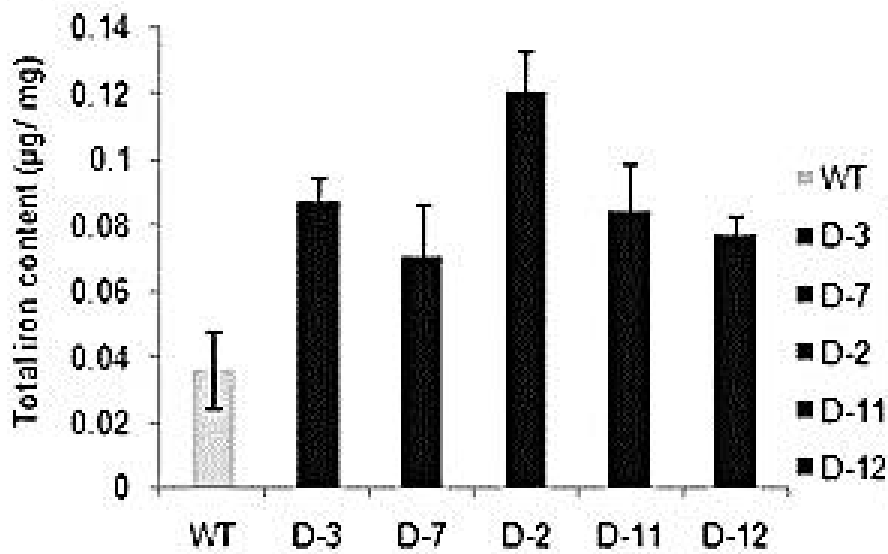


Fig. 3.1

(<http://www.nature.com/srep/2012/121210/srep00951/full/srep00951.html>)

- (a) Explain the term “transgenic plants”.

.....
[1]

- (b) With reference to Fig. 3.1, account for the effect of introducing the transgene into the crops.

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[2]

Further investigations were carried out to determine if *FvC5sdp* gene taken from different species of fungus would enable maize plants to take up greater quantities of iron.

- *FvC5sdp* cDNA genes were obtained from four species of fungus (*Flammulina rossica*, *Flammulina similis*, *Flammulina stratosa* and *Flammulina velutipes*). The cDNA genes were inserted into different plasmids.
- CaMV 35S promoter, a maize endosperm-specific promoter were also inserted into all of the plasmids.
- The four types of recombinant plasmids were then inserted into different cultures of maize cells.
- The iron content present in the maize plants was determined by atomic absorption spectroscopy.

The results are shown in Table 3.1.

Table 3.1

Source of <i>FvC5sdp</i> cDNA gene	Total Iron Content of Maize Plants / arbitrary units
<i>Flammulina rossica</i>	2
<i>Flammulina similis</i>	4
<i>Flammulina stratosa</i>	6
<i>Flammulina velutipes</i>	14

- (c) Explain how the *FvC5sdp* cDNA was produced from each of the fungus species.

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[2]

- (d) Suggest why the CaMV 35S promoter was inserted into the plasmids.

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[1]

- (e) State **one** environmental risk of transgenic crops and **one** health risk from consuming these crops.

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.....[2]

[Total: 8]

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4. *Pipistrellus* is a genus of bats in the family Vespertilionidae. Classifying bat species based on their evolutionary relationship is difficult because of the large diversity of species. Fig. 4.1 shows the phylogenetic tree of the four known bat species (*Pipistrell pipistrellus*, *Pipistrell pygmaeus*, *Pipistrell javanicus* and *Pipistrellus abramus*).

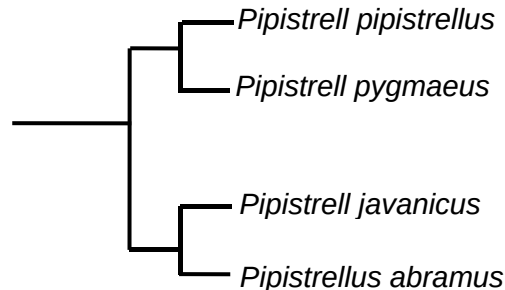


Fig. 4.1

Using this information and your own knowledge, design an experiment to show the molecular homology between four species of bats.

Your planning must be based on the assumption that you have been provided with the following equipment and materials.

- Tissue samples of the four bats species
- Pestle and mortar
- DNA extraction buffer solution
- Glass rods
- Microfuge tubes
- Centrifuge machine
- Restriction enzyme, HindIII
- Suitable reagents for polymerase chain reaction
- DNA primers
- Thermal cycler
- Agarose gel plate
- Suitable source of electrical current

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure,
- a description of the method used,
- relevant, clearly labelled diagram,
- relevant risk and precaution taken.
- The correct use of technical and scientific terms

[Total: 12]

[illegible]

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[illegible]

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5.

- (a)** Describe the use of restriction fragment length polymorphism analysis in creating a linkage map. [6]
- (b)** Describe the unique features of stem cells and explain normal functions of embryonic stem cells in a living organism. [6]
- (c)** Using named plant and animal examples, discuss the significance of genetic engineering in improving the quantity of food supply [8]

[Total: 20]

End of Paper